KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5083B | 5 Credits: 4 | Program Elective | Source-grounded

Laser and Optical

 To impart knowledge about the fundamentals of optics and opto-electronic devices.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5083B is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5083B.pdf from the uploaded official SITTTR syllabus archive.
Course code and title: preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Instrumentation Engineering
Course code	5083B
Course title	Laser and Optical
Semester	5 Credits: 4
Course category	Program Elective
Periods per week	4 (L:4 T:0 P:0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

15 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To impart knowledge about the fundamentals of optics and opto-electronic devices.
- To illustrate the applications of Lasers in various fields.
- To understand about fiber optic communication.
- To illustrate the principles of transducers in fiber optic sensors.

Course outcomes

CO	Official outcome	Study level
CO1	16 Understanding fundamentals of Laser Illustrate the classification and applications of	See official syllabus
CO2	14 Applying Explain the fundamentals of fiber optic	See official syllabus
CO3	communication systems and optoelectronic 15 Understanding components Illustrate the principles of transducers in fiber	See official syllabus
CO4	13 Applying optic sensors	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 2
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Describe the properties of light and fundamentals of Laser	4	As specified
Module 2	Illustrate the classification and applications of Laser	3	As specified
Module 3	optoelectronic components	4	As specified
Module 4	Illustrate the principles of transducers in fiber optic sensors	4	As specified

MODULE 1

Describe the properties of light and fundamentals of Laser

This module is presented from the official syllabus scope for Laser and Optical. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Describe the properties of light and fundamentals of Laser
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▾ refraction, dispersion, interference, diffraction 6 Understanding and polarization Describe Young’s double slit experiment for

refraction, dispersion, interference, diffraction 6 Understanding and polarization Describe Young’s double slit experiment for is an official topic in Module 1 of Laser and Optical. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify refraction, dispersion, interference, diffraction 6 Understanding and polarization Describe Young’s double slit experiment for using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, refraction, dispersion, interference, diffraction 6 understanding and polarization describe young’s double slit experiment for should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What refraction, dispersion, interference, diffraction 6 Understanding and polarization Describe Young’s double slit experiment for means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ refraction, dispersion, interference, diffraction 6 Understanding and polarization Describe Young’s double slit experiment for □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ 3 Understanding interference phenomenon Illustrate Newton’s ring experiment and its

3 Understanding interference phenomenon Illustrate Newton’s ring experiment and its is an official topic in Module 1 of Laser and Optical. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding interference phenomenon Illustrate Newton’s ring experiment and its using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding interference phenomenon illustrate newton’s ring experiment and its should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 3 Understanding interference phenomenon Illustrate Newton’s ring experiment and its means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

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Malayalam support: Module 1-□□ 3 Understanding interference phenomenon Illustrate Newton’s ring experiment and its □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ 3 Understanding application Explain the characteristics and principles of

3 Understanding application Explain the characteristics and principles of is an official topic in Module 1 of Laser and Optical. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding application Explain the characteristics and principles of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding application explain the characteristics and principles of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 3 Understanding application Explain the characteristics and principles of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 3 Understanding application Explain the characteristics and principles of □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ 4 Understanding Laser Phenomenon and laws of reflection - refraction - dispersion-interference - Young's double slit experiment - phase difference and path difference - constructive and destructive interference - conditions for bright and dark fringes - conditions for distribution of light intensities - Newton's Rings - Application of Newton's rings - diffraction - difference between interference and diffraction - polarization - Characteristics of laser beam - Interaction between matter and radiation - absorption - spontaneous emission - stimulated emission - Thermal equilibrium - Population inversion - Basic requirements for producing laser

4 Understanding Laser Phenomenon and laws of reflection - refraction - dispersion-interference - Young's double slit experiment - phase difference and path difference - constructive and destructive interference - conditions for bright and dark fringes - conditions for distribution of light intensities - Newton's Rings - Application of Newton's rings - diffraction - difference between interference and diffraction - polarization - Characteristics of laser beam - Interaction between matter and radiation - absorption - spontaneous emission - stimulated emission - Thermal equilibrium - Population inversion - Basic requirements for producing laser is an official topic in Module 1 of Laser and Optical. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding Laser Phenomenon and laws of reflection - refraction - dispersion-interference - Young's double slit experiment -

phase difference and path difference - constructive and destructive interference - conditions for bright and dark fringes - conditions for distribution of light intensities - Newton’s Rings - Application of Newton’s rings - diffraction - difference between interference and diffraction - polarization - Characteristics of laser beam - Interaction between matter and radiation - absorption - spontaneous emission - stimulated emission - Thermal equilibrium - Population inversion - Basic requirements for producing laser using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding laser phenomenon and laws of reflection - refraction - dispersion-interference - young’s double slit experiment - phase difference and path difference - constructive and destructive interference - conditions for bright and dark fringes - conditions for distribution of light intensities - newton’s rings - application of newton’s rings - diffraction - difference between interference and diffraction - polarization - characteristics of laser beam - interaction between matter and radiation - absorption - spontaneous emission - stimulated emission - thermal equilibrium - population inversion - basic requirements for producing laser should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding Laser Phenomenon and laws of reflection - refraction - dispersion-interference - Young’s double slit experiment - phase difference and path difference - constructive and destructive interference - conditions for bright and dark fringes - conditions for distribution of light intensities - Newton’s Rings - Application of Newton’s rings - diffraction - difference between interference and diffraction - polarization - Characteristics of laser beam - Interaction

Aspect	What to prepare
	between matter and radiation - absorption - spontaneous emission - stimulated emission - Thermal equilibrium - Population inversion - Basic requirements for producing laser means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 4 Understanding Laser Phenomenon and laws of reflection - refraction - dispersion-interference - Young’s double slit experiment - phase difference and path difference - constructive and destructive interference - conditions for bright and dark fringes - conditions for distribution of light intensities - Newton’s Rings - Application of Newton’s rings - diffraction - difference between interference and diffraction - polarization - Characteristics of laser beam - Interaction between matter and radiation - absorption - spontaneous emission - stimulated emission - Thermal equilibrium - Population inversion - Basic requirements for producing laser □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Illustrate the classification and applications of Laser

This module is presented from the official syllabus scope for Laser and Optical. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Illustrate the classification and applications of Laser
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ 8 Understanding Laser,CO2 Laser, Liquid dye Laser, Semiconductor Laser Identify the industrial, medical, scientific,

8 Understanding Laser,CO2 Laser, Liquid dye Laser, Semiconductor Laser Identify the industrial, medical, scientific, is an official topic in Module 2 of Laser and Optical. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 8 Understanding Laser,CO2 Laser, Liquid dye Laser, Semiconductor Laser Identify the industrial, medical, scientific, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 8 understanding laser,co2 laser, liquid dye laser, semiconductor laser identify the industrial, medical, scientific, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 8 Understanding Laser,CO2 Laser, Liquid dye Laser, Semiconductor Laser Identify the industrial, medical, scientific, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 8 Understanding Laser,CO2 Laser, Liquid dye Laser, Semiconductor Laser Identify the industrial, medical, scientific, □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▾ military and miscellaneous applications of 5 Applying laser Identify the application of Laser in

military and miscellaneous applications of 5 Applying laser Identify the application of Laser in is an official topic in Module 2 of Laser and Optical. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify military and miscellaneous applications of 5 Applying laser Identify the application of Laser in using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, military and miscellaneous applications of 5 applying laser identify the application of laser in should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What military and miscellaneous applications of 5 Applying laser Identify the application of Laser in means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ military and miscellaneous applications of 5 Applying laser Identify the application of Laser in □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ 1 Applying instrumentation area Classification of laser - Construction and Working of - Ruby laser - Nd - YAG laser - He - Ne Laser - Argon Ion Laser - CO2 Laser - Liquid dye Laser - Semiconductor Laser- Industrial Applications of laser (laser welding, cutting, drilling) - Medical applications of laser - Scientific applications of laser - Military Applications of laser - Miscellaneous applications of laser(laser printer optical disc recording and reading, holography, construction of hologram and reconstruction of image) - Laser application in Instrumentation (displacement measurement - Laser Doppler velocity meter) Explain the fundamentals of fiber optic communication systems and

1 Applying instrumentation area Classification of laser - Construction and Working of - Ruby laser - Nd - YAG laser - He - Ne Laser - Argon Ion Laser - CO2 Laser - Liquid dye Laser - Semiconductor Laser- Industrial Applications of laser (laser welding, cutting, drilling) - Medical applications of laser - Scientific applications of laser - Military Applications of laser - Miscellaneous applications of laser(laser printer optical disc recording and reading, holography, construction of hologram and reconstruction of image) - Laser application in Instrumentation (displacement measurement - Laser Doppler velocity meter) Explain the fundamentals of fiber optic communication systems and is an official topic in Module 2 of Laser and Optical. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Applying instrumentation area Classification of laser - Construction and Working of - Ruby laser - Nd - YAG laser - He - Ne Laser - Argon Ion Laser - CO2 Laser - Liquid dye Laser - Semiconductor Laser- Industrial Applications of laser (laser welding, cutting, drilling) - Medical applications of laser - Scientific applications of laser - Military Applications of laser - Miscellaneous applications of laser(laser printer optical disc recording and reading, holography, construction of hologram and reconstruction of image) - Laser application in Instrumentation (displacement measurement - Laser Doppler velocity meter) Explain the fundamentals of fiber optic communication systems and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 applying instrumentation area classification of laser - construction and working of - ruby laser - nd - yag laser - he - ne laser - argon ion laser - co2 laser - liquid dye laser - semiconductor laser- industrial applications of laser (laser welding, cutting, drilling) - medical applications of laser - scientific applications of laser - military applications of laser - miscellaneous applications of laser(laser printer optical disc recording and reading, holography, construction of hologram and reconstruction of image) - laser application in instrumentation (displacement measurement - laser doppler velocity meter) explain the fundamentals of fiber optic communication systems and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Applying instrumentation area Classification of laser - Construction and Working of - Ruby laser - Nd - YAG laser - He - Ne Laser - Argon Ion Laser - CO2 Laser - Liquid dye Laser - Semiconductor Laser-

Aspect	What to prepare
	Industrial Applications of laser (laser welding, cutting, drilling) - Medical applications of laser - Scientific applications of laser - Military Applications of laser - Miscellaneous applications of laser(laser printer optical disc recording and reading, holography, construction of hologram and reconstruction of image) - Laser application in Instrumentation (displacement measurement - Laser Doppler velocity meter) Explain the fundamentals of fiber optic communication systems and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] 1 Applying instrumentation area Classification of laser - Construction and Working of - Ruby laser - Nd - YAG laser - He - Ne Laser - Argon Ion Laser - CO2 Laser - Liquid dye Laser - Semiconductor Laser- Industrial Applications of laser (laser welding, cutting, drilling) - Medical applications of laser - Scientific applications of laser - Military Applications of laser - Miscellaneous applications of laser(laser printer optical disc recording and reading, holography, construction of hologram and reconstruction of image) - Laser application in Instrumentation (displacement measurement - Laser Doppler velocity meter) Explain the fundamentals of fiber optic communication systems and [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

optoelectronic components

This module is presented from the official syllabus scope for Laser and Optical. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: optoelectronic components
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 2 Understanding operation of optical fiber Explain the modes of operation of optical

2 Understanding operation of optical fiber Explain the modes of operation of optical is an official topic in Module 3 of Laser and Optical. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding operation of optical fiber Explain the modes of operation of optical using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding operation of optical fiber explain the modes of operation of optical should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 2 Understanding operation of optical fiber Explain the modes of operation of optical means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 2 Understanding operation of optical fiber Explain the modes of operation of optical □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ 2 Understanding fiber Explain the block diagram and advantages

2 Understanding fiber Explain the block diagram and advantages is an official topic in Module 3 of Laser and Optical. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding fiber Explain the block diagram and advantages using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding fiber explain the block diagram and advantages should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 2 Understanding fiber Explain the block diagram and advantages means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 2 Understanding fiber Explain the block diagram and advantages □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ 4 Understanding of fiber optic communication Describe the construction and working of optoelectronic components(photo diode-

4 Understanding of fiber optic communication Describe the construction and working of optoelectronic components(photo diode- is an official topic in Module 3 of Laser and Optical. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding of fiber optic communication Describe the construction and working of optoelectronic components(photo diode- using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding of fiber optic communication describe the construction and working of optoelectronic components(photo diode- should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding of fiber optic communication Describe the construction and working of optoelectronic components(photo diode- means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 4 Understanding of fiber optic communication Describe the construction and working of optoelectronic components(photo diode- □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□□.

▼ 7 Understanding PiN diode- Avalanche photo Diode-photo transistor- LED) Principle of Optical Fibers - critical angle - total internal reflection - optical fiber construction - acceptance angle and Numerical aperture - modes of operation of optical fibers - Single mode step index fiber - multi mode step index fiber - multimode graded index fiber - block diagram of Fiber optic Communication System - Advantages of fiber optic communication - Construction and working of photo diode - PiN diode - Avalanche Photo Diode-photo transistor - LED - principle of operation - LED drive circuits.

7 Understanding PiN diode- Avalanche photo Diode-photo transistor- LED) Principle of Optical Fibers - critical angle - total internal reflection - optical fiber construction - acceptance angle and Numerical aperture - modes of operation of optical fibers - Single mode step index fiber - multi mode step index fiber - multimode graded index fiber - block diagram of Fiber optic Communication System - Advantages of fiber optic communication - Construction and working of photo diode - PiN diode - Avalanche Photo Diode-photo transistor - LED - principle of operation - LED drive circuits. is an official topic in Module 3 of Laser and Optical. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 7 Understanding PiN diode- Avalanche photo Diode-photo transistor- LED) Principle of Optical Fibers - critical angle - total internal reflection - optical fiber construction - acceptance angle and Numerical aperture - modes of operation of optical fibers - Single mode step index fiber - multi mode step index fiber - multimode graded index fiber - block diagram of Fiber optic Communication System - Advantages of fiber optic communication - Construction and working of photo diode - PiN diode - Avalanche Photo Diode-photo transistor - LED - principle of operation - LED drive circuits. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 7 understanding pin diode- avalanche photo diode-photo transistor- led) principle of optical fibers - critical angle - total internal reflection - optical fiber construction - acceptance angle and numerical aperture - modes of operation of optical fibers - single mode step index fiber - multi mode step index fiber - multimode graded index fiber - block diagram of fiber optic communication system - advantages of fiber optic communication - construction and working of photo diode - pin diode - avalanche photo diode-photo transistor - led - principle of operation - led drive circuits. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 7 Understanding PiN diode- Avalanche photo Diode-photo transistor- LED) Principle of Optical Fibers - critical angle - total internal reflection - optical fiber construction - acceptance angle and Numerical aperture - modes of operation of optical fibers - Single mode step index fiber - multi mode step index fiber - multimode graded index fiber - block diagram of Fiber optic Communication System - Advantages of fiber optic communication - Construction and working of photo diode - PiN diode - Avalanche Photo Diode-photo transistor - LED - principle of operation - LED drive circuits. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 7 Understanding PiN diode- Avalanche photo Diode-photo transistor- LED) Principle of Optical Fibers - critical angle - total internal reflection - optical fiber construction - acceptance angle and Numerical aperture - modes of operation of optical fibers - Single mode step index fiber - multi mode step index fiber - multimode graded index fiber - block diagram of Fiber optic Communication System - Advantages of fiber optic communication - Construction and working of photo diode - PiN diode - Avalanche Photo Diode-photo transistor - LED - principle of operation - LED drive circuits. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Illustrate the principles of transducers in fiber optic sensors

This module is presented from the official syllabus scope for Laser and Optical. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Illustrate the principles of transducers in fiber optic sensors
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 2 Understanding sensor. Differentiate between passive and active

2 Understanding sensor. Differentiate between passive and active is an official topic in Module 4 of Laser and Optical. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding sensor. Differentiate between passive and active using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding sensor. differentiate between passive and active should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 2 Understanding sensor. Differentiate between passive and active means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 2 Understanding sensor. Differentiate between passive and active □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ 1 Understanding optical fiber sensors Application of fiber optic sensor (displacement, pressure, temperature, force,

1 Understanding optical fiber sensors Application of fiber optic sensor (displacement, pressure, temperature, force, is an official topic in Module 4 of Laser and Optical. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding optical fiber sensors Application of fiber optic sensor (displacement, pressure, temperature, force, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding optical fiber sensors application of fiber optic sensor (displacement, pressure, temperature, force, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding optical fiber sensors Application of fiber optic sensor (displacement, pressure, temperature, force, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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Malayalam support: Module 4-□□ 1 Understanding optical fiber sensors Application of fiber optic sensor (displacement, pressure, temperature, force, □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ 8 Applying torque, strain, level, flow and humidity measurement) Describe the advantages of fiber optic

8 Applying torque, strain, level, flow and humidity measurement) Describe the advantages of fiber optic is an official topic in Module 4 of Laser and Optical. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 8 Applying torque, strain, level, flow and humidity measurement) Describe the advantages of fiber optic using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 8 applying torque, strain, level, flow and humidity measurement) describe the advantages of fiber optic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 8 Applying torque, strain, level, flow and humidity measurement) Describe the advantages of fiber optic means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 8 Applying torque, strain, level, flow and humidity measurement) Describe the advantages of fiber optic □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ 2 Understanding sensor General structure of fiber optic sensor-passive and active optical fiber sensors- Displacement sensor- Pressure sensor- Temperature sensor- Force sensor- Torque sensor- Strain sensor - Level sensor- Flow sensor-humidity sensor-Advantages of fiber optic sensor. Text / Reference: T/R Book Title/Author S. Nagabhushana and N. Sathyanarayana, Lasers and Optical Instrumentation, I K R1 International Publishing House R2 N. Subrahmanyan & Brijlal, A Text Book of Optics,S Chand publications R3 B.L. Theraja, Engineering Physics, S Chand publications R4 Keiser, Optical Fiber Communication System, TMH R5 K. Thyagaraja & Ghatak , Laser theory & application, Springer R6 T.L Singal, Optical fiber communications, Cambridge university press Online Resources: Sl.No Website Link 1 https://www.youtube.com/watch?v=yfawFJCRDSE 2 https://www.youtube.com/watch?v=gIGOXNlvMsg 3 https://www.youtube.com/watch?v=Z2Kme6ozj7U 4 https://www.youtube.com/watch?v=Gsj44ObhL24 5 http://www.powershow.com/view/348ef- 6 https://www.docslides.com/cheryl-pisano/fiber-optical-sensor 7 http://www.circuitstoday.com/how-a-led-works-light-emitting-diode-working 8 https://en.wikipedia.org/wiki/List_of_laser_applications

2 Understanding sensor General structure of fiber optic sensor-passive and active optical fiber sensors- Displacement sensor- Pressure sensor- Temperature sensor- Force sensor- Torque sensor- Strain sensor - Level sensor- Flow sensor-humidity sensor-Advantages of fiber optic sensor. Text / Reference: T/R Book Title/Author S. Nagabhushana and N. Sathyanarayana, Lasers and Optical Instrumentation, I K R1 International Publishing House R2 N. Subrahmanyan & Brijlal, A Text Book of Optics,S Chand publications R3 B.L. Theraja, Engineering Physics, S Chand publications R4 Keiser, Optical Fiber Communication System, TMH R5 K. Thyagaraja & Ghatak , Laser theory & application, Springer R6 T.L Singal, Optical fiber communications, Cambridge university press Online Resources: Sl.No Website Link 1 https://www.youtube.com/watch?v=yfawFJCRDSE 2 https://www.youtube.com/watch?v=gIGOXNlvMsg 3

https://www.youtube.com/watch?v=Z2Kme6ozj7U 4
https://www.youtube.com/watch?v=Gsj44ObhL24 5
http://www.powershow.com/view/348ef- 6
https://www.docslides.com/cheryl-pisano/fiber-optical-sensor 7
http://www.circuitstoday.com/how-a-led-works-light-emitting-diode-working 8 https://en.wikipedia.org/wiki/List_of_laser_applications is an official topic in Module 4 of Laser and Optical. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding sensor General structure of fiber optic sensor-passive and active optical fiber sensors- Displacement sensor- Pressure sensor- Temperature sensor- Force sensor- Torque sensor- Strain sensor - Level sensor- Flow sensor-humidity sensor-Advantages of fiber optic sensor. Text / Reference: T/R Book Title/Author S. Nagabhushana and N. Sathyanarayana, Lasers and Optical Instrumentation, I K R1 International Publishing House R2 N. Subrahmanyan & Brijlal, A Text Book of Optics,S Chand publications R3 B.L. Theraja, Engineering Physics, S Chand publications R4 Keiser, Optical Fiber Communication System, TMH R5 K. Thyagaraja & Ghatak , Laser theory & application, Springer R6 T.L Singal, Optical fiber communications, Cambridge university press Online Resources: Sl.No Website Link 1 https://www.youtube.com/watch?v=yfawFJCRDSE 2 https://www.youtube.com/watch?v=gIGOXNlvMsg 3 https://www.youtube.com/watch?v=Z2Kme6ozj7U 4 https://www.youtube.com/watch?v=Gsj44ObhL24 5 http://www.powershow.com/view/348ef- 6 https://www.docslides.com/cheryl-pisano/fiber-optical-sensor 7 http://www.circuitstoday.com/how-a-led-works-light-emitting-diode-working 8 https://en.wikipedia.org/wiki/List_of_laser_applications using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding sensor general structure of fiber optic sensor-passive and active optical fiber sensors- displacement sensor- pressure sensor- temperature sensor- force sensor- torque sensor- strain sensor - level sensor- flow sensor-humidity sensor-advantages of fiber optic sensor. text / reference: t/r book title/author s. nagabhushana and n. sathyanarayana, lasers and optical instrumentation, i k r1 international publishing house r2 n. subrahmanyan & brijlal, a text book of optics,s chand publications r3 b.l. theraja, engineering physics, s chand publications r4 keiser, optical fiber communication system, tmh r5 k. thyagaraja & ghatak , laser theory & application, springer r6 t.l singal, optical fiber communications, cambridge university press online resources: sl.no website link 1 https://www.youtube.com/watch?v=yfawfjcrdse 2 https://www.youtube.com/watch?v=gigoxnlvmsg 3 https://www.youtube.com/watch?v=z2kme6ozj7u 4 https://www.youtube.com/watch?v=gsj44obhl24 5 http://www.powershow.com/view/348ef- 6 https://www.docslides.com/cheryl-pisano/fiber-optical-sensor 7 http://www.circuitstoday.com/how-a-led-works-light-emitting-diode-working 8 https://en.wikipedia.org/wiki/list_of_laser_applications should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding sensor General structure of fiber optic sensor-passive and active optical fiber sensors- Displacement sensor- Pressure sensor- Temperature sensor- Force sensor- Torque sensor- Strain sensor - Level sensor- Flow sensor-humidity sensor-Advantages of fiber optic

Aspect	What to prepare
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Study lens	Principle → components or stages → result → application
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Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 2 Understanding sensor General structure of fiber optic sensor-passive and active optical fiber sensors-Displacement sensor- Pressure sensor- Temperature sensor- Force sensor-Torque sensor- Strain sensor - Level sensor- Flow sensor-humidity sensor-Advantages of fiber optic sensor. Text / Reference: T/R Book Title/Author S. Nagabhushana and N. Sathyanarayana, Lasers and Optical Instrumentation, I K R1 International Publishing House R2 N. Subrahmanyan & Brijlal, A Text

Book of Optics,S Chand publications R3 B.L. Theraja, Engineering Physics, S Chand publications R4 Keiser, Optical Fiber Communication System, TMH R5 K. Thyagaraja & Ghatak , Laser theory & application, Springer R6 T.L Singal, Optical fiber communications, Cambridge university press Online Resources: Sl.No Website Link 1 https://www.youtube.com/watch?v=yfawFJCRDSE 2 https://www.youtube.com/watch?v=gIGOXNlvMsg 3 https://www.youtube.com/watch?v=Z2Kme6ozj7U 4 https://www.youtube.com/watch?v=Gsj44ObhL24 5 http://www.powershow.com/view/348ef- 6 https://www.docslides.com/cheryl-pisano/fiber-optical-sensor 7 http://www.circuitstoday.com/how-a-led-works-light-emitting-diode-working 8 https://en.wikipedia.org/wiki/List_of_laser_applications □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle →
Components/steps →
Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure
→ Observation → Result →
Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
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Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

▶ **Define or explain refraction, dispersion, interference, diffraction 6 Understanding and polarization Describe Young's double slit experiment for. (3 marks)**

▶ **Define or explain 3 Understanding interference phenomenon Illustrate Newton's ring experiment and its. (3 marks)**

▶ **Define or explain 3 Understanding application Explain the characteristics and principles of. (3 marks)**

▶ **Define or explain 4 Understanding Laser Phenomenon and laws of reflection - refraction - dispersion-interference - Young's double slit experiment - phase difference and path difference - constructive and destructive interference - conditions for bright and dark fringes - conditions for distribution of light intensities - Newton's Rings - Application of Newton's rings - diffraction - difference between interference and diffraction - polarization - Characteristics of laser beam - Interaction between matter and radiation - absorption - spontaneous emission - stimulated emission - Thermal equilibrium - Population inversion - Basic requirements for producing laser. (3 marks)**

Module 2

▶ **Define or explain 8 Understanding Laser,CO2 Laser, Liquid dye Laser, Semiconductor Laser Identify the industrial, medical, scientific,. (3**

marks)

▶ **Define or explain military and miscellaneous applications of 5 Applying laser Identify the application of Laser in. (3 marks)**

▶ **Define or explain 1 Applying instrumentation area Classification of laser - Construction and Working of - Ruby laser - Nd - YAG laser - He - Ne Laser - Argon Ion Laser - CO2 Laser - Liquid dye Laser - Semiconductor Laser- Industrial Applications of laser (laser welding, cutting, drilling) - Medical applications of laser - Scientific applications of laser - Military Applications of laser - Miscellaneous applications of laser(laser printer optical disc recording and reading, holography, construction of hologram and reconstruction of image) - Laser application in Instrumentation (displacement measurement - Laser Doppler velocity meter) Explain the fundamentals of fiber optic communication systems and. (3 marks)**

Module 3

▶ **Define or explain 2 Understanding operation of optical fiber Explain the modes of operation of optical. (3 marks)**

▶ **Define or explain 2 Understanding fiber Explain the block diagram and advantages. (3 marks)**

▶ **Define or explain 4 Understanding of fiber optic communication Describe the construction and working of optoelectronic components(photo diode-. (3 marks)**

▶ **Define or explain 7 Understanding PiN diode- Avalanche photo Diode- photo transistor- LED) Principle of Optical Fibers - critical angle - total internal reflection - optical fiber construction - acceptance angle and Numerical aperture - modes of operation of optical fibers - Single mode step index fiber - multi mode step index fiber - multimode graded index**

fiber - block diagram of Fiber optic Communication System - Advantages of fiber optic communication - Construction and working of photo diode - PiN diode - Avalanche Photo Diode-photo transistor - LED - principle of operation - LED drive circuits.. (3 marks)

Module 4

▶ **Define or explain 2 Understanding sensor. Differentiate between passive and active. (3 marks)**

▶ **Define or explain 1 Understanding optical fiber sensors Application of fiber optic sensor (displacement, pressure, temperature, force,. (3 marks)**

▶ **Define or explain 8 Applying torque, strain, level, flow and humidity measurement) Describe the advantages of fiber optic. (3 marks)**

▶ **Define or explain 2 Understanding sensor General structure of fiber optic sensor-passive and active optical fiber sensors- Displacement sensor- Pressure sensor- Temperature sensor- Force sensor- Torque sensor- Strain sensor - Level sensor- Flow sensor-humidity sensor- Advantages of fiber optic sensor. Text / Reference: T/R Book Title/Author S. Nagabhushana and N. Sathyanarayana, Lasers and Optical Instrumentation, I K R1 International Publishing House R2 N. Subrahmanyan & Brijlal, A Text Book of Optics,S Chand publications R3 B.L. Theraja, Engineering Physics, S Chand publications R4 Keiser, Optical Fiber Communication System, TMH R5 K. Thyagaraja & Ghatak , Laser theory & application, Springer R6 T.L Singal, Optical fiber communications, Cambridge university press Online Resources: Sl.No Website Link 1 https://www.youtube.com/watch?v=yfawFJCRDSE 2 https://www.youtube.com/watch?v=gIGOXNlvMsg 3 https://www.youtube.com/watch?v=Z2Kme6ozj7U 4 https://www.youtube.com/watch?v=Gsj44ObhL24 5 http://www.powershow.com/view/348ef- 6 https://www.docslides.com/cheryl-pisano/fiber-optical-sensor 7**

http://www.circuitstoday.com/how-a-led-works-light-emitting-diode-working 8 https://en.wikipedia.org/wiki/List_of_laser_applications. (3 marks)

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **refraction, dispersion, interference, diffraction 6 Understanding and polarization Describe Young’s double slit experiment for**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **8 Understanding Laser,CO2 Laser, Liquid dye Laser, Semiconductor Laser Identify the industrial, medical, scientific,**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **2 Understanding operation of optical fiber Explain the modes of operation of optical**.

Module 4

1-mark definition: State the meaning of **2 Understanding sensor. Differentiate between passive and active**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link https://www.youtube.com/watch?v=yfawFJCRDSE https://www.youtube.com/watch?v=gIGOXNlvMsg https://www.youtube.com/watch?v=Z2Kme6ozj7U https://www.youtube.com/watch?v=Gsj44ObhL24 http://www.powershow.com/view/348ef-
- https://www.docslides.com/cheryl-pisano/fiber-optical-sensor
- http://www.circuitstoday.com/how-a-led-works-light-emitting-diode-working
- https://en.wikipedia.org/wiki/List_of_laser_applications

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